

Laser Specifications

1. Laser Specifications

1.1. Optical Characteristics

- Operation Mode: CW / Modulated
- Polarization: Random / Un-polarized
- Nominal Maximum Output Power: 1000 W
- Emission Wavelength: 1070 nm at 1000 W output power
- Emission Linewidth: < 5 nm at 1000 W output power
- Short-term Power Instability: <2% rms at 10 kHz to 20 MHz frequency range, 1000 W power
- Long-term Power Instability: Less than ± 3 % at 1000 W and 4 hours operation
- Switching ON/OFF Time: 30-50 μ s
- Output power: 1000 W
- Power Modulation Rate: up to 50 kHz at 1000 W

1.2. Optical Output

- Beam Quality (M2): < 1.1 mm
- Fiber diameter: < 15 μ m
- Output Fiber Termination: QBH-compatible connector
- Beam Divergence: > 85 mrad and <115 mrad at full angle, 86% level
- Delivery Cable Bending Radius: >80 mm

1.3. Electrical characteristics

- Operating voltage: 200-240 VAC, single phase, 50/60 Hz
- Max Power Consumption: 3500 W

1.4. General characteristics

- Cabinet Dimensions: height < 225 mm (5 rack mount units)
- Width/Mounting: 19" rack-mountable
- Weight: <85 kg
- Control: Full operation over RS-232, Ethernet, or touch-screen